KNOWBRIDGE

AI-POWERED PROJECT KNOWLEDGE HANDOVER PLATFORM • HACKATHON ROUND 2 SUBMISSION

Team Name	Tech Titan	**Participation**	Individual Participant (Aman Prajapati)
Round	Round 2 – The Build	**Live URL**	https://knowbridge-production.up.railway.app/
Repository	https://github.com/Amanprajapati247/KnowBridge.git	**Lead Account**	lead@knowbridge.io (PW: DemoLead123!)
Incoming Account	incoming@knowbridge.io (PW: Incoming123!)	**Automated Tests**	5/5 Pytest Suites Passing (100%)

1. Executive Summary & Round 2 Objective

When an employee, developer, or student lead departs a project, critical institutional knowledge disappears with them. Standard documentation tools fail because key context is scattered across disparate notes, specs, and chat logs, with zero verification of what is current versus obsolete. In Round 1, Team Tech Titan designed the 5-step handover architecture. In Round 2 (The Build), this has been engineered into a **functional, judge-ready, cloud-deployed SaaS platform** live on Railway at **https://knowbridge-production.up.railway.app/**. The system ingests multi-format documents, segments passages with coordinate tracking, extracts 10 categories of structured intelligence, detects cross-document contradictions, and enforces strict role-based handover views.

2. Core End-to-End Workflow Demonstration

KnowBridge successfully executes the complete five-stage knowledge handover workflow without broken steps:

Step	Capability	Demonstrated Implementation on Live Deployment
1. Connect	Multi-Format Ingestion	Drag-and-drop ingestion of PDF, DOCX, and TXT files. PyMuPDF and python-docx extract text into segmented passages preserving filename, page numbers, and byte offsets. Raw files downloadable directly from UI.
2. Reconstruct	Structured Extraction	AI pipeline extracts 10 operational categories: Decisions, Tasks, Owners, Deadlines, Risks, Blockers, Dependencies, Questions, Milestones, and Permitted Contacts. Validated with strict Pydantic schemas.
3. Verify	Evidence Grounding & Conflicts	Deterministic passage linking with VERIFIED, UNCERTAIN, and CONFLICT states. Discrepancies between sources (e.g. conflicting target launch dates) are highlighted side-by-side without guessing.
4. Review	Lead Studio & Redaction	Project leads resolve conflicts, edit claims, redact sensitive operational details, and officially approve the 14-section handover report.
5. Continue	Role-Based Continuation	Incoming members access only approved handovers. Backend RBAC scrubs redacted items and completely blocks access to unapproved draft reports and restricted files.

3. Technical Depth, Reliability & Architecture

KnowBridge is built with a production-grade enterprise stack prioritizing stability, zero-hallucination grounding, and clean separation of concerns.

Component	Technology	Architectural Responsibility
Frontend Client	React 18, TypeScript, Vite, Tailwind	Enterprise SPA dashboard, Recharts visual analytics, interactive Evidence Modal.
Backend API	FastAPI (Python 3.12+), Pydantic v2	Asynchronous REST endpoints, strict request/response validation schemas.
Database & ORM	SQLAlchemy 2.0, SQLite (Postgres ready)	Relational models: Project, Document, Passage, Claim, Evidence, Handover, AuditLog.
Document Extraction	PyMuPDF (fitz), python-docx	Extracts text passages with exact page coordinates, paragraph boundaries, and byte offsets.
Authentication	PyJWT, Passlib (Bcrypt)	Stateless JSON Web Tokens with role claims (PROJECT_LEAD vs. INCOMING_MEMBER).
Deployment Engine	Multi-Stage Dockerfile, Railway Cloud	Unified single-port deployment: FastAPI serves REST API and client-side SPA bundle.

Grounding & Anti-Hallucination Guardrails:

- **Strict Passage Anchoring:** Every extracted claim is linked via foreign keys to a concrete passage ID. Statements cannot exist without supporting evidence.
- **No Silent Winner Selection:** Conflicting dates or owners are retained with status 'CONFLICT', preventing dangerous automated assumptions.
- **Absence Disclosure:** Missing task assignees or deadlines are explicitly marked as *'Not found in available sources'* rather than fabricated.
- **Deterministic Fallback:** Evaluators can test the system without external API keys via a robust rule-based NLP extraction pipeline.

4. Progress Against Round 1 Plan & Technical Evolution

Team Tech Titan has delivered 100% of the proposed Round 1 scope while implementing key architectural enhancements:

Planned Capability (Round 1)	Status	Implementation & Evolution in Round 2
File Ingestion (PDF, DOCX, TXT)	**Delivered**	Built with PyMuPDF and python-docx; added 25MB validation and direct file download.
Traceable Citations & Offsets	**Delivered**	Interactive modal showing exact text passage, source filename, page number, and offset.
Contradiction Detection	**Delivered**	Pairwise entity conflict detection with interactive resolution UI for project leads.
14-Section Handover Report	**Delivered**	Generates complete handover structure including decisions, tasks, risks, and next steps.
Lead Review & Redaction	**Delivered**	In-place claim editing, one-click redaction toggles, and official sign-off flow.
Role-Based Access Control	**Delivered**	Backend-enforced JWT permissions isolating incoming members from draft reports.
Real-World Project Seeding	**Delivered**	Pre-fed with 3 complete enterprise projects, 10 documents, and 48 extracted claims.
Live Cloud Deployment	**Delivered**	Unified single-port Docker container running live on Railway with zero CORS friction.

5. Verification, Testing & Live Multi-Project Data

The codebase was verified through 5 automated test suites executed with 100% pass rates, as well as end-to-end integration verification on the live Railway production domain:

- **Automated Pytest Suite (5/5 Passed):** Health check, role authorization, document extraction, conflict engine, and handover approval.
- **Project 1 – College Event Management System (CEMS):** 5 documents, 23 claims, cross-document launch date contradiction flagged, Handover v1.0 **APPROVED**.
- **Project 2 – Campus Autonomous Shuttle Service:** 3 documents, 15 claims, LiDAR sensor obstacle risk flagged, Handover v1.0 **DRAFT**.
- **Project 3 – CareLink Student Wellness Platform:** 2 documents, 10 claims, HIPAA compliance rules verified, Handover v1.0 **APPROVED**.

6. Three Substantial Commitments for Round 3

In compliance with the Round 2 brief, Team Tech Titan commits to three major product improvements for Round 3:

Commitment	Proposed Round 3 Improvement	Impact & Real-World Value
1. Live Git & Task Connectors	Native API ingestion for **GitHub** repositories (PRs, commits) and **Jira / Linear** issue boards.	Captures architectural decisions from PR reviews and issue status transitions in real time, eliminating manual file uploads.
2. Semantic Hybrid RAG	Vector embeddings with **pgvector** combined with lexical BM25 search.	Enables sub-second knowledge retrieval across 10,000+ pages and supports interactive Q&A; over the approved handover.
3. Cryptographic Audit Trail	SHA-256 content hashing, version diff viewer (v1.0 vs. v2.0), and incoming sign-off acknowledgment receipt.	Provides a legally binding compliance audit trail verifying that incoming members reviewed and accepted project responsibilities.

7. Usability, User Experience & SaaS Polish

KnowBridge was built to commercial SaaS standards. The interface features a clean navy (#0f172a) and teal (#0d9488) visual palette, high-contrast verification badges, Recharts visual breakdowns, and a dedicated workspace navigation layout. All demo credentials, evaluation shortcuts, and test tags were completely removed, presenting an authentic enterprise application ready for immediate enterprise evaluation.

8. Fair Participation & Build Window Disclosures

This project was developed during the official hackathon build window by individual participant **Aman Prajapati** under team name **Tech Titan**. Disclosed open-source libraries: React 18, Vite, Tailwind CSS, FastAPI, SQLAlchemy, PyMuPDF, python-docx, Passlib, PyJWT. AI coding assistance was utilized during development for boilerplate generation and test scaffolding. All architectural logic, domain schemas, extraction algorithms, and design choices were directed and verified by the author.